

Sea Food Supply Chain Management Using Blockchain

¹N. Nasurudeen Ahamed, ²Karthikeyan P, ³S.P.Anandaraj, ⁴Vignesh R

^{1,2,3,4} Department of Computer Science and Engineering, Presidency University, Bengaluru, India

¹nasurmece@gmail.com, ²nrmkarthi@gmail.com, ³anandsofttech@gmail.com, ⁴hananyaresearch@gmail.com

Abstract - Blockchain is a distributed open ledger that is used to record the transaction across many computers. Blockchain technology can be applied in any domain such as banking, healthcare, real estate, travel, food, and supply chain. Blockchain allows us to have a distributed peer to peer networks, so it provides security and privacy. Example: Bitcoin. To enhance the Supply chain management using in the blockchain using some unique tags (Example: RFID/NFC/QR Code) tags describe the information about products and track the products. This paper discusses the blockchain in supply chain management to eliminate the disadvantages of the supply chain in the life cycle. In order to provide the solution of supply chain management to using these unique tags to get all details for manufacturing date/expiry date, etc., from the manufacturer side so no can alter this tag for the manufacturer to build a trustworthy of wholesaler/retailer/consumer for using the special tags.

Keywords: Blockchain, Peer to Peer Network, Ledger, Supply Chain, RFID, NFC, QR Code.

I. INTRODUCTION

The Blockchain is pervasive technology for its decentralized infrastructure. The blockchain contains a block of information. It is not possible to change block information or backdate the information. Blockchain technologies classified into the following types: 1. Public Blockchain 2. Private Blockchain 3. Consortium Blockchain. Public Blockchain: All Public Peer can join one another and also have equal rights (Example: Read, Write, and Execute) in the public node. Private Blockchain: Only Authorized Private Peer can access in the network. The access to the node in the private peer is only open to the individual node. Consortium Blockchain: Only Authorized Team can only access and join this blockchain, all operation in the node must obey to the access. Blockchain Technology mainly focused on the transaction in the business network model.

Many facets of today's markets are likely to be impacted by large scale adoption of the blockchain method. In the Blockchain era, we entered the Supply Chain Management(SCM)[1][2].supply chain is the primary concern of business area while transporting among the producer and consumer[3][4]. Many

companies across the globe to manage the supply chain from source to destiny. Supply chain management[5] is the flow of products travel from producer to consumer in the right and systematic way[6]. That is why the majority of companies nowadays approach this concept. There are significant advantages of this supply chain management is: Customer Fulfilment, Identify the problems, and cost efficiency[7].

However, the problem of this sector is the delay of delivered products between the parties as well as the other issues. In this approach, we take Raw material products as an example. In this proposal, we are going to implement the two aspects while traveling the products from producer to consumer. First, we reduce the delay while the delivery of goods[8][9]. That is to use the smart tags wholesaler/retailer/consumer know the product information and delivery details can collect. Second, we improve the trustworthy in one more step ahead at the customer level.

That is wholesaler/retailer/consumer get the product from producer using the smart tag to know the detail about particular product (example: Company Name, Manufacturing Date, etc.)

The major contribution of the paper is summarized as follows:

1. Steered detail related work on Sea food Supply chain management.
2. Designed Smart tags for sea food supply chain using Blockchain.

The rest of this paper is organized as follows: in Section 2, we discussed the related work of sea food supply chain management using in blockchain. In Section 3, we described the proposed sea food supply chain management using in blockchain. In Section 4 we concluded with future work.

II. RELATED WORK

Blockchain is a distributed open ledger that is used to record the transaction across many computers. Blockchain technology can be applied in any domain

such as banking, healthcare, real estate, travel, food, and supply chain. In this section, we examine some related existing works. (1) Supply Chain using NFC (2) RFID in Supply Chain (3) DL-Tags

A. Supply Chain using NFC

Naif Alzahrani [10] proposed to use the NFC Tag track and trace the product. Each product using NFC Tag, Securing, and storing the product. Each NFC tag [11] has Unique Tag ID (TID).

Nikos et al. [12] Discussed the NFC tag using object tracking in supply chain management. However, this paper describes not only tracing also NFC enabled handsets when picking a set of tagged objects. That is, a handheld device will download the list of objects to be picked.

Xi Zhang et al. [13] discussed the use of NFC Tag to monitoring the Products at various temperatures. That is, this not belongs to the consumer. It applies to the manufacturer to smart packing the product use of NFC Tag.

All existing work we described in the above used NFC in the supply chain management to use this tag to monitor the product from the manufacturer side or consumer side. That is tracking the product item using the NFC tag.

B. RFID in Supply Chain

Oh-Keun Ha et al. [14] described in the supply chain using RFID tag to implement the food industry monitoring transportation [15], sales, and administration.

Muhammad Qasim Saeed et al. [16] Proposed the consumer to detect the counterfeit product using RFID Tag.

The above described existing related work in the supply chain management using RFID Tag [17][11] to similar in the NFC tag to monitoring the transportation [18] the goods from producer to consumer and also find the counterfeit the products before to deliver the customer. That is the customer to use the RFID tag to detect the product, whether counterfeit or not.

C. DL-Tags

Federico Matteo et al. [19] proposed to prevent the product duplication and track the product from origin to destination in the supply chain life cycle. That is some of the smart tags (i.e., NFC/RFID/QR CODE) used in the supply chain management enhanced by the Internet of Things (IoT).

The above described the related work is defined to use the Distributed Ledger Technology [20] (DLT) offers a decentralized and privacy-preserving in during the product's life cycle.

Table I: Difference between NFC / RFID / QR Code

SPECIAL TAGS			
CONSTRAINTS	NFC	RFID	QR CODE
Network Type	Point – Point	Point – Point	Point – Point
Communication	Bi-directional	Uni-directional	Uni-directional
Range	More than 0.2 m	Up to 100 m	1 cm
Cost	Medium	High	Low
Security	High	High	Low

III. Sea Food Supply Chain Management Using Blockchain Technology

This paper proposed to implement the new supply chain management in the blockchain technology. Through this implementation of this new model. The following Figure 1 represents the new model for supply chain management to use in the blockchain. During the Supply Chain Management, we projected to implement three unique tags to combine in the supply chain management as follows:

1. NFC (Near Field Communication)
2. RFID (Radio Frequency Identification)
3. QR Code. These are the three unique tags implemented in supply chain management. Here, for example, we take the Seafood supply chain in the blockchain. Regarding this seafood supply chain, we get the raw material (Seafood) from fisherman, and its send to the appropriate Retailer / Consumer using some unique tag to enable in these supply chains.

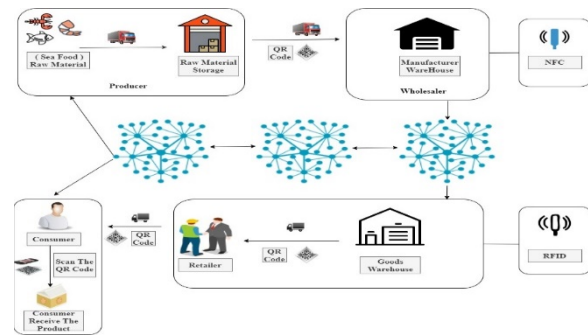


Fig.1. Sea Food Supply Chain Using Special Tags.

The above Fig 1 classified into two categories one is The supplier part of another one is the Retailer / Consumer part. In the Supplier Part, we get the raw material (Sea Food) from the fisherman and send to the raw material production/storage to keeping in the warehouse to ready to supply the goods to retailer/customer before that we have to generate the QR code for that particular product. So in these parts, much transparency to using in the blockchain. Now we designed to enable the special tag in the supplier part that is NFC.

Regarding this, NFC (Near Field Communication) is a very short-range communication device under the IEEE 802.11 standards. These NFC enabled in the supplier part when it is ready to send to the customer part.

The purpose of NFC enabled in the supplier part it is beneficial to communicate one and two – way communication between endpoints within the concise range communication if any material those who damaged while to send the goods to the consumer immediately we can replace before its send to the consumer. Retailer/ Customer part in the supply chain management to enabled RFID technology. The purpose of this technology mainly used in commerce. It is handy for customers to track their assets.

The purpose of QR code generated from the warehouse to reach the appropriate consumer means it is for consumer satisfaction. After reaching the product to the consumer. He/she check out the product via mobile to check the good's origin and date of the particular food. Fig 2.illustrate, the seafood supply chain, using in the blockchain. The producer/wholesaler/retailer/consumer is connected in the blockchain, and through the blockchain payment, and product details are trace via some special tags (Example: RFID / NFC /QR Code).

The writing research and the overview infer that blockchain innovation shows potential for the fish business to increment the straightforwardness and detectability of the store network. Be that as it may, it must be noticed that this innovation is generally new and has numerous mechanical and financial difficulties to survive to understand its maximum capacity. On an administration and industry level, partners are associated with an appropriated record, which will expand the reliability and straightforwardness of the business. Items can be effectively followed and followed, and exercises are completed by type of agreement. As fish is a nutritious and significant wellspring of nourishment, the moves made to invigorate maintainability through guidelines, such as standards and certificates, can be based on blockchain innovation for more prominent precision and unwavering quality. On an organization level,

organizations in the catch fishery industry could improve their efficiency and limit cost by digitizing their documentation process with parties legitimately engaged with their center business exercises. Also, blockchain innovation might bolster roundabout economy standards and comprehensive business models and increment the estimation of side-effects by improved straightforwardness and reliability.

IV CONCLUSION AND FUTURE WORK

This paper defines a new blockchain method to enhance the current supply chain, management model. In supply chain commodities to connect the people from various locations. So we have to implement the distributed ledger in the supply chain. Distribute ledger in the above seafood supply chain all transactions and transportation. It is very confident that no one can attack. Moreover, we proposed some unique tags we included from the producer to reach the consumer. Advantages of special tag all people from producer to consumer side to get the details about the product that is trustworthy of product manufacturing.

Future of this research, we have to implement the IoT smart sensor to include in the supply chain commodities using blockchain and secure the Products using Blockchain and Artificial Intelligence.

References

- [1] R. Casado-Vara, J. Prieto, F. De La Prieta, and J. M. Corchado, "How blockchain improves the supply chain: Case study alimentary supply chain," *Procedia Comput. Sci.*, vol. 134, pp. 393–398, 2018.
- [2] S. Tönnissen and F. Teuteberg, "Analysing the impact of blockchain-technology for operations and supply chain management: An explanatory model drawn from multiple case studies," *Int. J. Inf. Manage.*, no. April, pp. 0–1, 2019.
- [3] J. Al-Jaroodi and N. Mohamed, "Blockchain in Industries: A Survey," *IEEE Access*, vol. 7, pp. 36500–36515, 2019.
- [4] M. Montecchi, K. Plangger, and M. Etter, "It's real, trust me! Establishing supply chain provenance using blockchain," *Bus. Horiz.*, vol. 62, no. 3, pp. 283–293, 2019.
- [5] R. Azzi, R. K. Chamoun, and M. Sokhn, "The power of a blockchain-based supply chain," *Comput. Ind. Eng.*, vol. 135, no. August 2018, pp. 582–592, 2019.
- [6] S. A. A., "Blockchain Ready Manufacturing Supply Chain Using Distributed Ledger," *Int. J. Res. Eng. Technol.*, vol. 05, no. 09, pp. 1–10, 2016.
- [7] Y. Cui and H. Idota, "Improving supply chain resilience with establishing a decentralized information sharing mechanism," *ACM Int. Conf. Proceeding Ser.*, pp. 1–7, 2018.
- [8] H. R. Hasan and K. Salah, "Blockchain-Based Proof of Delivery of Physical Assets with Single and Multiple Transporters," *IEEE Access*, vol. 6, no. 8, pp. 46781–46793, 2018.
- [9] G. Perboli, S. Musso, and M. Rosano, "Blockchain in

Logistics and Supply Chain: A Lean Approach for Designing Real-World Use Cases,” *IEEE Access*, vol. 6, pp. 62018–62028, 2018.

- [10] N. Alzahrani and N. Bulusu, “Block-supply chain: A new anti-counterfeiting supply chain using NFC and blockchain,” *CRYBLOCK 2018 - Proc. 1st Work. Cryptocurrencies Blockchains Distrib. Syst. Part MobiSys 2018*, pp. 30–35, 2018.
- [11] Y.-J. Tu and S. Piramuthu, “On Addressing RFID/NFC-based Relay Attacks: An Overview,” *Decis. Support Syst.*, 2019.
- [12] N. Kefalakis, N. Leontiadis, J. Soldatos, K. Gama, and D. Donsez, “Supply chain management and NFC picking demonstrations using the AspireRfid middleware platform,” *Proc. ACM/IFIP/USENIX Middlew. 2008 Conf. Companion*, pp. 66–69, 2008.
- [13] X. Zhang, X. Shan, and J. Wei, “Hybrid flexible smart temperature tag with NFC technology for smart packaging,” *2017 IEEE 19th Electron. Packag. Technol. Conf. EPTC 2017*, vol. 2018-Febru, pp. 1–5, 2018.
- [14] O. K. Ha, Y. S. Song, K. Y. Chung, K. D. Lee, and D. Park, “Relation model describing the effects of introducing RFID in the supply chain: Evidence from the food and beverage industry in South Korea,” *Pers. Ubiquitous Comput.*, vol. 18, no. 3, pp. 553–561, 2014.
- [15] Y. P. Tsang, K. L. Choy, C. H. Wu, G. T. S. Ho, and H. Y. Lam, “Blockchain-Driven IoT for Food Traceability With an Integrated Consensus Mechanism,” *IEEE Access*, vol. 7, pp. 129000–129017, 2019.
- [16] M. Q. Saeed, Z. Bilal, and C. D. Walter, “An NFC based consumer-level counterfeit detection framework,” *2013 11th Annu. Conf. Privacy, Secur. Trust. PST 2013*, pp. 135–142, 2013.
- [17] S. Mondal, K. P. Wijewardena, S. Karuppuswami, N. Kriti, D. Kumar, and P. Chahal, “Blockchain inspired RFID-based information architecture for food supply chain,” *IEEE Internet Things J.*, vol. 6, no. 3, pp. 5803–5813, 2019.
- [18] P. Helo and A. H. M. Shamsuzzoha, “Real-time supply chain—A blockchain architecture for project deliveries,” *Robot. Comput. Integr. Manuf.*, vol. 63, no. November 2019, p. 101909, 2020.
- [19] F. M. Bencic, P. Skocir, and I. P. Zarko, “DL-Tags: DLT and Smart Tags for Decentralized, Privacy-Preserving, and Verifiable Supply Chain Management,” *IEEE Access*, vol. 7, pp. 46198–46209, 2019.
- [20] S. K. Lo *et al.*, “Analysis of Blockchain Solutions for IoT: A Systematic Literature Review,” *IEEE Access*, vol. 7, pp. 58822–58835, 2019.